



Exploratory Data Analysis (EDA) Summary Report

1. Introduction

This report aims to explore Geldium's customer dataset to uncover patterns and risk factors associated with loan delinquency. The goal is to assess data quality, identify predictive variables, and prepare the dataset for modelling by addressing missing values and inconsistencies. Insights from this analysis will guide feature selection and model design for delinquency prediction.

2. Dataset Overview

The dataset contains customer-level financial and behavioural data, including credit scores, income, loan balances, and monthly payment history.

Key dataset attributes:

- **Number of records:** 500+ customer entries
- **Key variables:**
 - **Credit_Utilization:** Ratio of credit used to available credit
 - **Missed_Payments:** Count of missed payments in recent months
 - **Debt_to_Income_Ratio:** Financial stress indicator
 - **Credit_Score:** Standard creditworthiness metric
 - **Monthly Payment Status (Month_1 to Month_6):** Behavioural history
 - **Delinquent_Account:** Binary target variable
- **Data types:**
- **Numerical:** Age, Income, Credit_Score, Loan_Balance, Debt_to_Income_Ratio
- **Categorical:** Employment_Status, Credit_Card_Type, Location, Monthly Payment Status

Anomalies & inconsistencies:

- Some Credit_Utilization values exceed 1.0, which is unrealistic.
- Employment_Status has inconsistent labels (e.g., "EMP", "employed", "Employed").
- A few records show zero loan balance with active missed payments.

3. Missing Data Analysis

Key missing data findings:

- **Credit_Utilization:** Missing or invalid values (e.g., >1.0)
- **Income:** Missing for some entries
- **Monthly Payment Status:** Missing or inconsistent across months

Missing data treatment:

Variable	Treatment Method	Justification
Credit_Utilization	Impute (Median)	Median reduces impact of outliers and preserves distribution.
Income	Synthetic Generation	Generated using age and employment status to maintain realism.
Monthly Payment Status	Impute (Mode)	Mode retains most common behaviour and supports categorical modelling.

4. Key Findings and Risk Indicators

Correlations observed:

- High Credit_Utilization (>0.6) correlates with increased missed payments.
- Debt_to_Income_Ratio > 0.4 is associated with higher delinquency rates.
- Low Credit_Score (<500) strongly predicts delinquency.

Unexpected anomalies:

- High-income individuals with frequent missed payments suggest behavioural risk.
- Young customers with high credit scores but poor payment history may indicate thin credit files.
- Retired individuals with high loan balances show mixed repayment behaviour.

High-risk indicators:

Indicator	Importance
Credit_Utilization > 0.6	Indicates financial overextension and repayment stress.
Missed_Payments ≥ 4	Direct behavioural signal of delinquency risk.
Debt_to_Income_Ratio > 0.4	Suggests poor affordability and high default probability.
Low Credit_Score (<500)	Reflects poor credit history and repayment behaviour.
Unemployed Status	Associated with lower repayment capacity and higher missed payments.

5. AI & GenAI Usage

Generative AI tools were used to accelerate data review, imputation strategy design, and pattern detection. These tools helped surface anomalies and prioritize predictive features.

Example AI prompts used:

- *"Summarize key patterns in the dataset and identify anomalies."*
- *"Suggest an imputation strategy for missing income values based on industry best practices."*
- *"Identify top variables most likely to predict delinquency."*

6. Conclusion & Next Steps

The dataset contains rich behavioural and financial data suitable for delinquency prediction. Key indicators such as credit utilization, missed payments, and debt-to-income ratio show strong predictive potential. However, data quality issues—especially inconsistent labels and unrealistic values—must be addressed before modelling.

Next steps include:

- Finalizing data cleaning and imputation
- Feature engineering (e.g., payment trend scores)
- Building baseline predictive models (e.g., logistic regression, decision trees)
- Validating model performance and fairness across customer segments